

Data Analyst Project

Facebook Content Monetization Revenue Prediction and Analysis

Muhammad Farel Daffa
July 2026

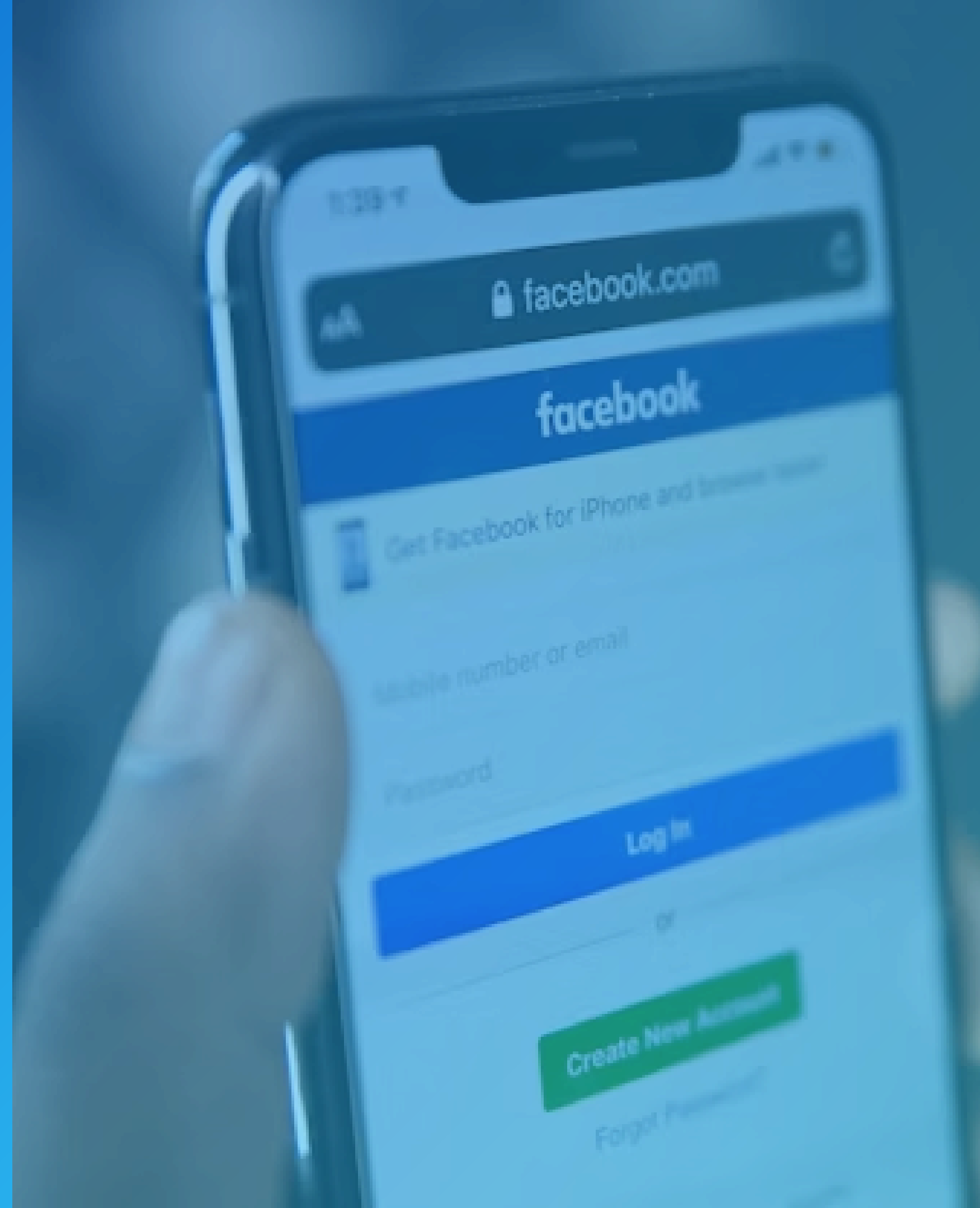
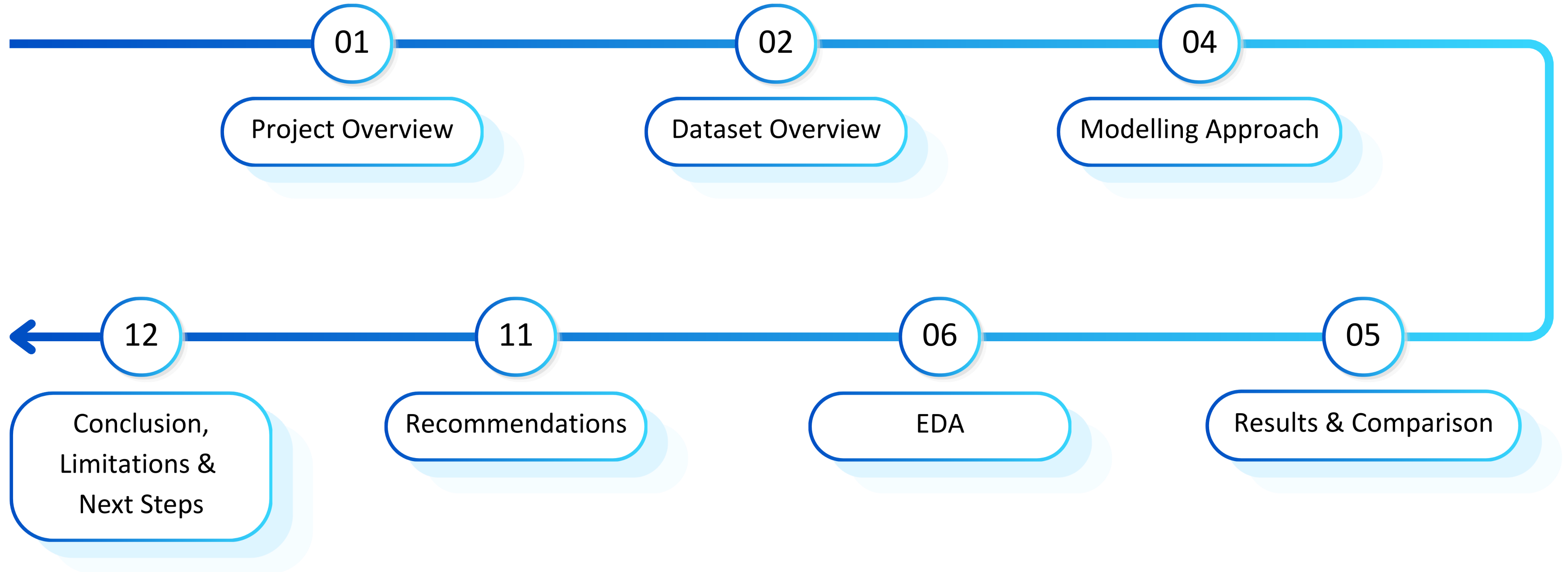


Table of Contents



Understanding the business problem, objectives, and analytical framework

Business Understanding

As competitors' Facebook monetization revenue is not publicly disclosed, machine learning models are developed to estimate revenue using available performance metrics. These estimates competitive analysis and help identify opportunities to improve Pikiran Rakyat's Facebook monetization strategy.

Objectives

- 01 Predict competitors' Facebook monetization revenue using machine learning models based on publicly available performance metrics.
- 02 Identify the key performance metrics that drive Facebook monetization revenue.
- 03 Analyze revenue patterns across media profiles and over time to uncover monetization strategies.

Framework

Data Understanding

Obtain and explore Facebook data from Meta Business Suite and SocialInsider.

Data Preparation

Process, transform, clean, standardize, and encoded the raw dataset into an modelling-ready format.

Modelling

Split 80%/20% dataset and train Linear Regression, Random Forest, and XGBoost.

Evaluation

Compare R^2 , MAE, and RMSE.

EDA

Identify key performance metrics that drive revenue and revenue patterns across media profiles.

Deployment

Recommendations for Facebook monetization strategy.

Dataset Overview



4,020

Total Posts



53

Total Variables



8

Media Profiles



2

Data Sources

Overview	Pikiran Rakyat	Competitors
 Period	1 May–7 June	1–7 June
 Rows	763	3,257
 Columns	35	18
 Missing Values	5,901	5,799
 Duplicate Rows	0	0
 Source	Meta Business Suite	SocialInsider

 The EDA will be conducted using data from the first week of June only. However, the May dataset from Pikiran Rakyat was required for model training to predict competitors' revenue.

Dataset Pre-Processing

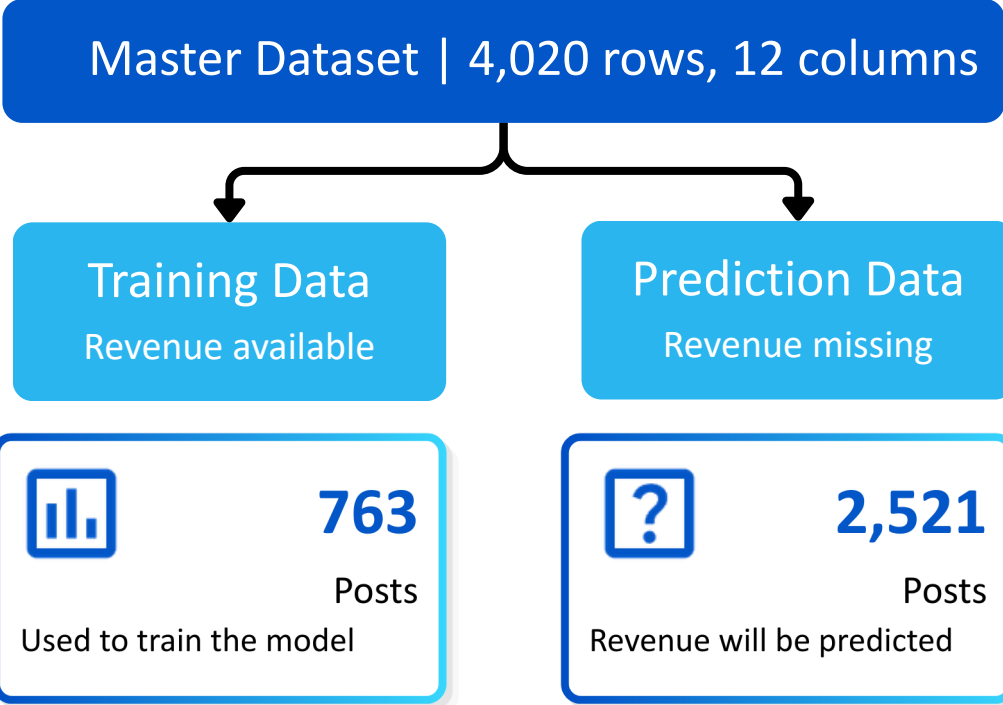


Preparing the dataset and features to predict missing Revenue values for competitor posts

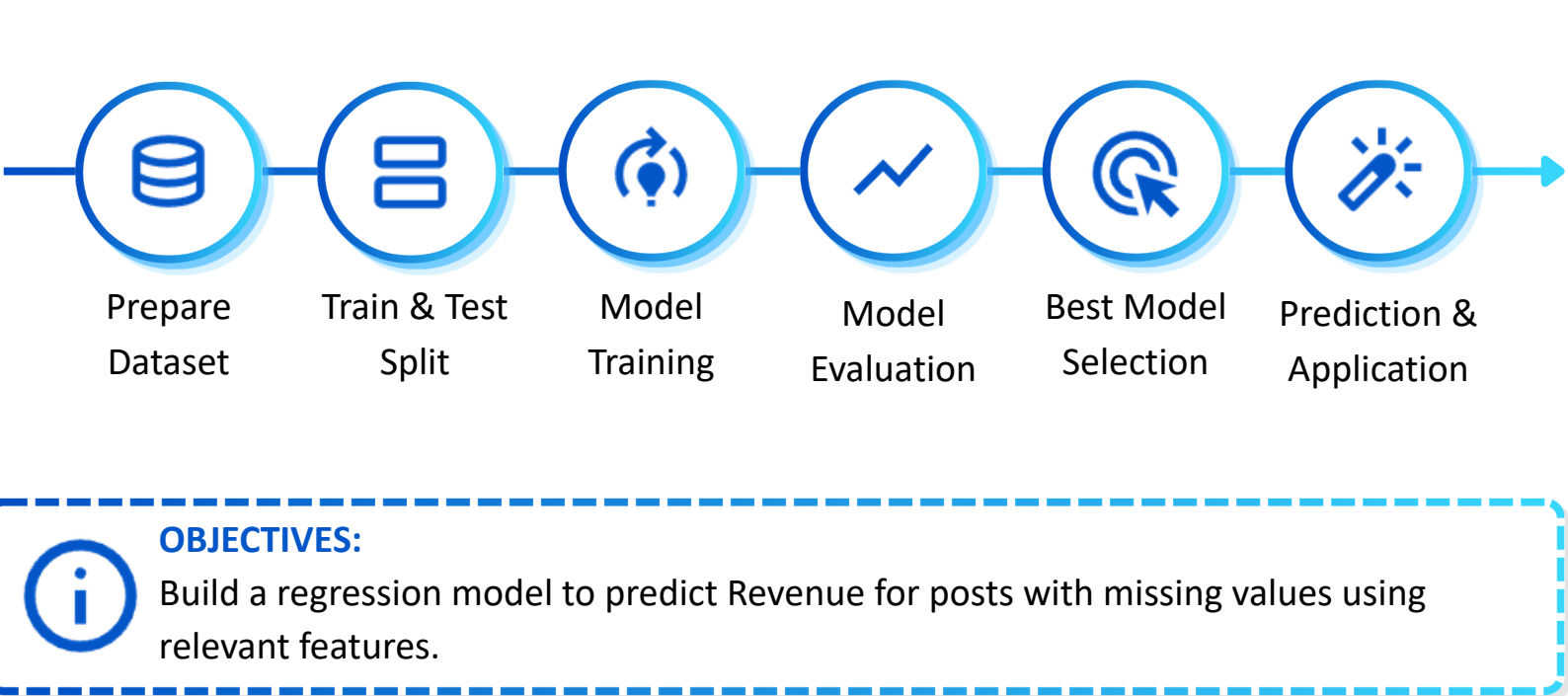
Why Revenue Prediction?

-  **2,521** competitor posts have missing Revenue values.
-  Revenue estimation was required to enable competitor benchmarking and analysis.
-  Missing revenue values were predicted using machine learning before analysis.
-  **TARGET VARIABLE:** Revenue (Log-transformed)

Data Split for Reach Prediction



Modelling Workflow



Feature Engineering



PROFILE INFORMATION

- Encoded Profile (e.g. 0: Antaranews , 1: Detik, 2: Kompas, etc.)

ENCODING



CONTENT FEATURES

- Encoded Post Type (e.g. 0: Link, 1: Photo, etc.)
- Word Count
- Hashtag Count

DERIVED + ENCODING



POSTING TIME FEATURES

- Posting Hour
- Encoded Day of the Week (e.g. 0: Friday, 1: Monday, etc.)

DERIVED + ENCODING



METRICS FEATURES

- Engagement
- Impressions
- Reach
- Reactions
- Comments
- Shares

LOG TRANSFORMATION

Train & Test Split

Train Set (80%)

610 Posts
Used to train the model

Test Set (20%)

153 Posts
Used to evaluate the model

RANDOM STATE: 42

Output

2,521 posts of Competitors were filled with predicted Revenue values

Modelling Workflow



1. Prepare Dataset
Cleaned dataset with engineered features and encoded variables.



2. Train & Test Split
Split data into training (80%) and testing (20%).



3. Model Training
Train multiple models using the training dataset.



4. Model Evaluation
Evaluate models on test set using multiple metrics.



5. Best Model Selection
Select the best performing model for prediction.



6. Prediction & Application
Generate predictions for missing values and downstream analysis

Models Evaluated



Linear Regression

A linear model that assumes a linear relationship between features and target variable.

 **STRENGTHS**

- Simple, fast, and easy to interpret
- Works well when relationships are linear
- Provides clear insights into feature impact

 **LIMITATIONS**

- Assumes linearity between features & target
- Sensitive to outliers
- May underperform on complex patterns



Random Forest

An ensemble of decision trees using bagging technique to improve accuracy and reduce overfitting.

 **STRENGTHS**

- Captures non-linear relationships
- Robust to outliers and noise
- Provides feature importance

 **LIMITATIONS**

- Less interpretable than linear models
- Can be computationally intensive
- Performance depends on parameter tuning



XGBoost

A gradient boosting algorithm that builds trees sequentially to correct errors from previous tree.

 **STRENGTHS**

- High predictive performance
- Handles complex patterns effectively
- Built-in regularization to reduce overfitting

 **LIMITATIONS**

- More sensitive to hyperparameters
- Requires more computational resources
- Less interpretable than linear models

Evaluation Metrics

R²

Measures the propotion of the variance explained by the model (0–1).

Higher is better

Root Mean Squared Error (RMSE)

Measures the average magnitude of prediction erros. Penalizes large erros more.

Lower is better

Mean Absolute Error (MAE)

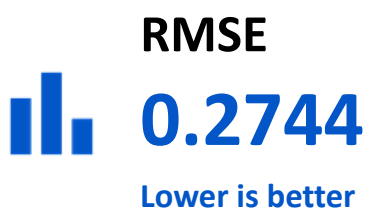
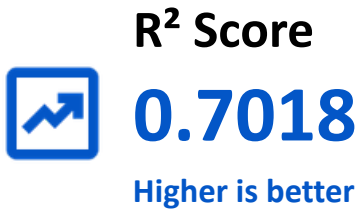
Measures the average absolute difference between actual and predicted values.

Lower is better

Revenue Prediction Results & Comparison

XGBoost was selected as the best model for predicting Revenue (Log-transformed)

Evaluation Metrics (Test Set)



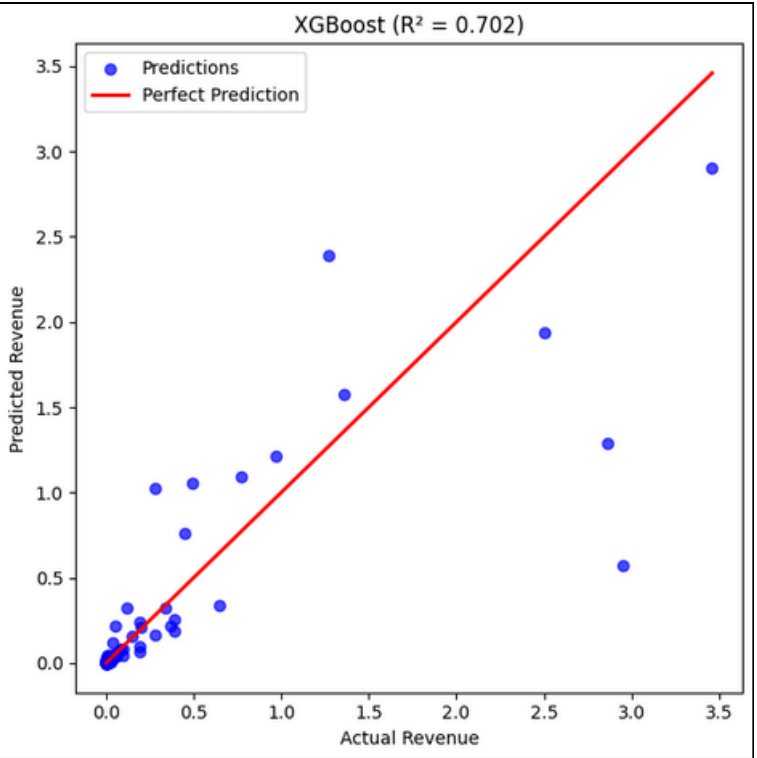
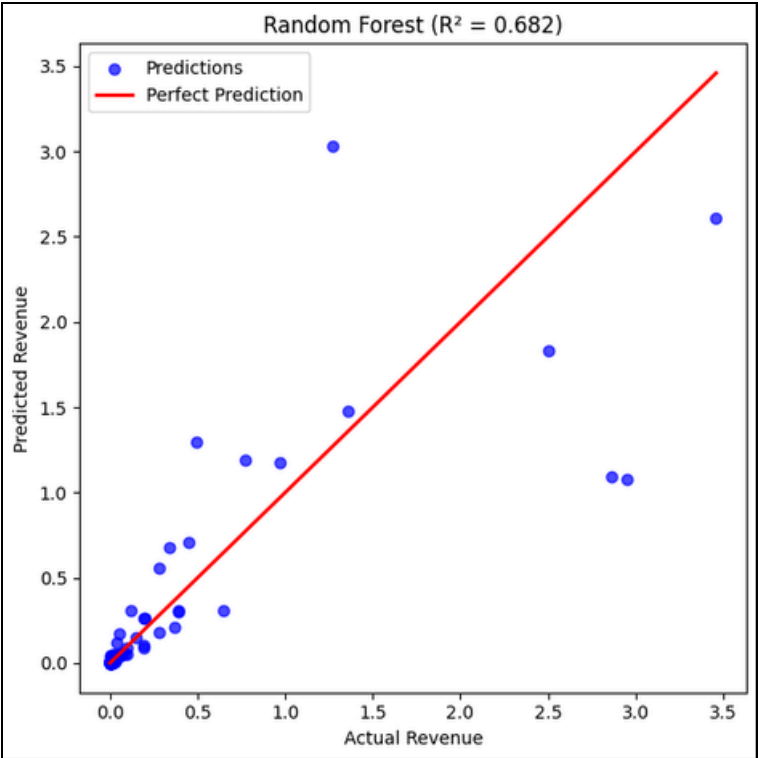
BEST MODEL
XGBoost

MAE 0.0728
RMSE 0.2744
R² 0.7018

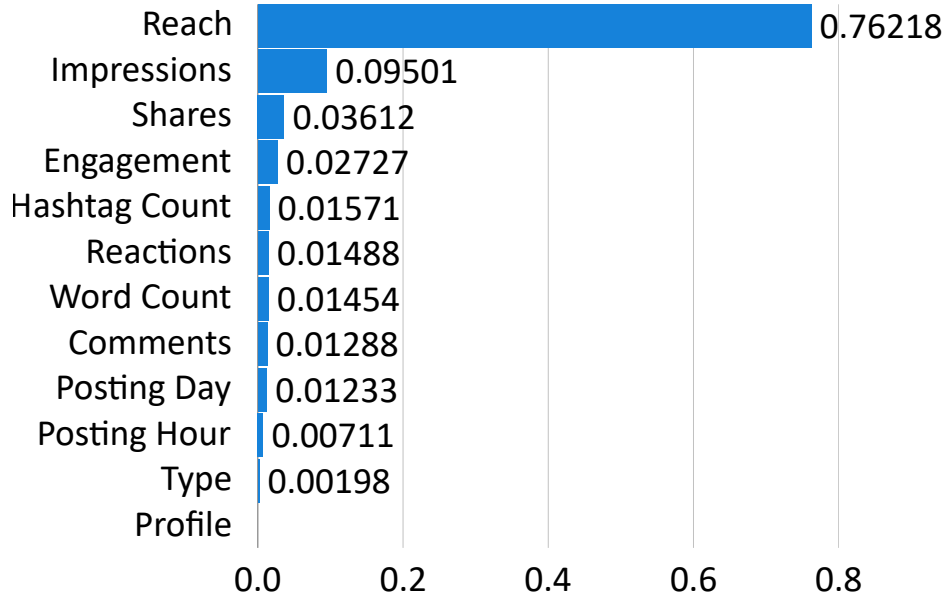
Model Comparison (Test Set)

Model	MAE	RMSE	R ²
 Linear Regression	0.181161	0.385890	0.409985
 Random Forest	0.076570	0.283328	0.681936
 XGBoost	0.072836	0.274354	0.701766

Actual vs Predict of Two Best Models



Feature Importance



Feature Importance Explanation

- Reach (Log)**
The most important feature, contributing 76.22%. Revenue was strongly driven by the amount of reach achieved by a post.
- Impressions (Log)**
Higher impressions increased the opportunity to generate monetization.
- Shares and Engagement (Log)**
Shares helped expand distribution and may lead to higher monetization. Engagement reflect audience interactions can improve content visibility and monetization potential.
- Content & Temporal Feature**
Word Count, Hashtag Count, Posting Day, Hour, and Content Type also contributed, but to a smaller extent.



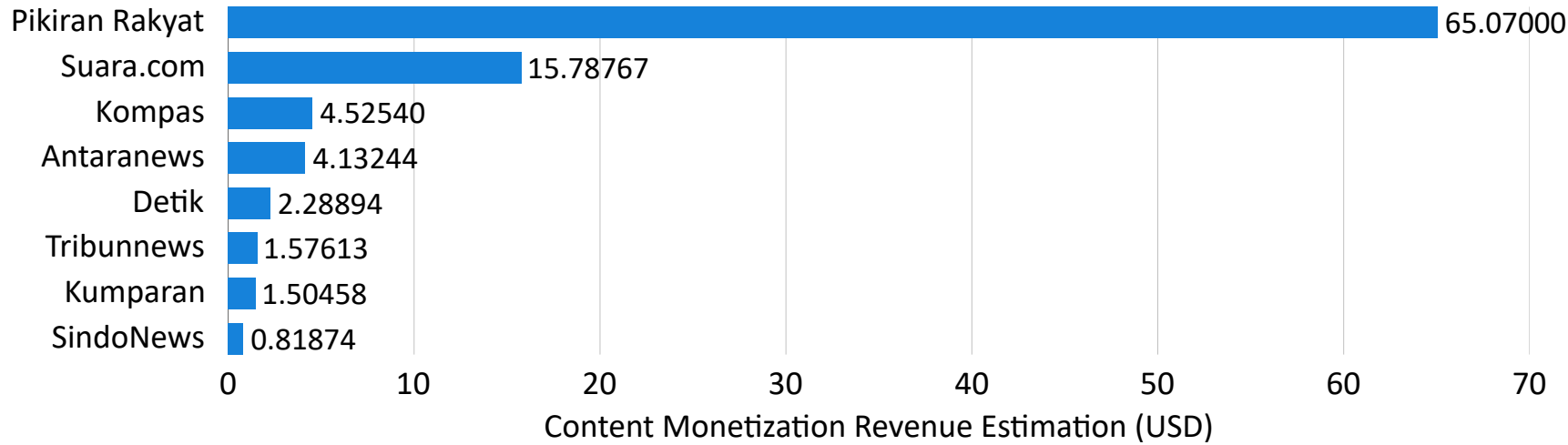
Key Insights

- XGBoost achieved the best performance across all metrics with R² of 0.7018, indicating that 70.18% of the variance in revenue was explained by the model.
- The model showed good predictive ability with low MAE (0.0728) and RMSE (0.2744) on the test set.
- Reach (log) was the most dominant feature, emphasizing the wider reach drives higher monetization revenue, followed by Impressions (log) and engagement-related metrics (log).
- The model was reliable and was used to predict missing Revenue values for competitors.

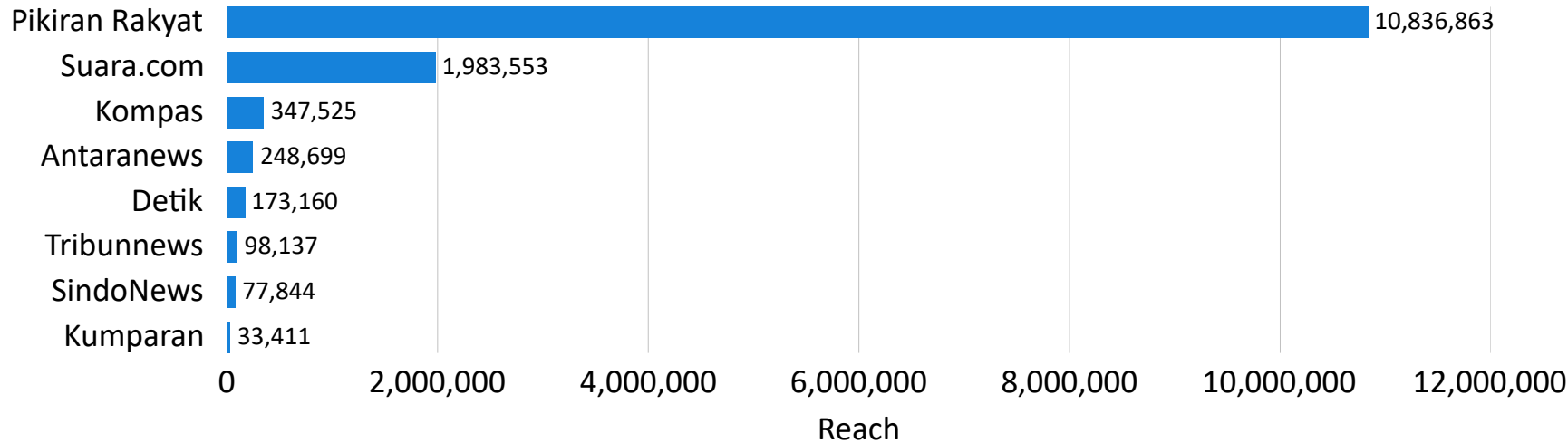
 XGBoost was selected as the final model for Revenue prediction.

Benchmarking overall Facebook performance across media profiles using aggregated metrics

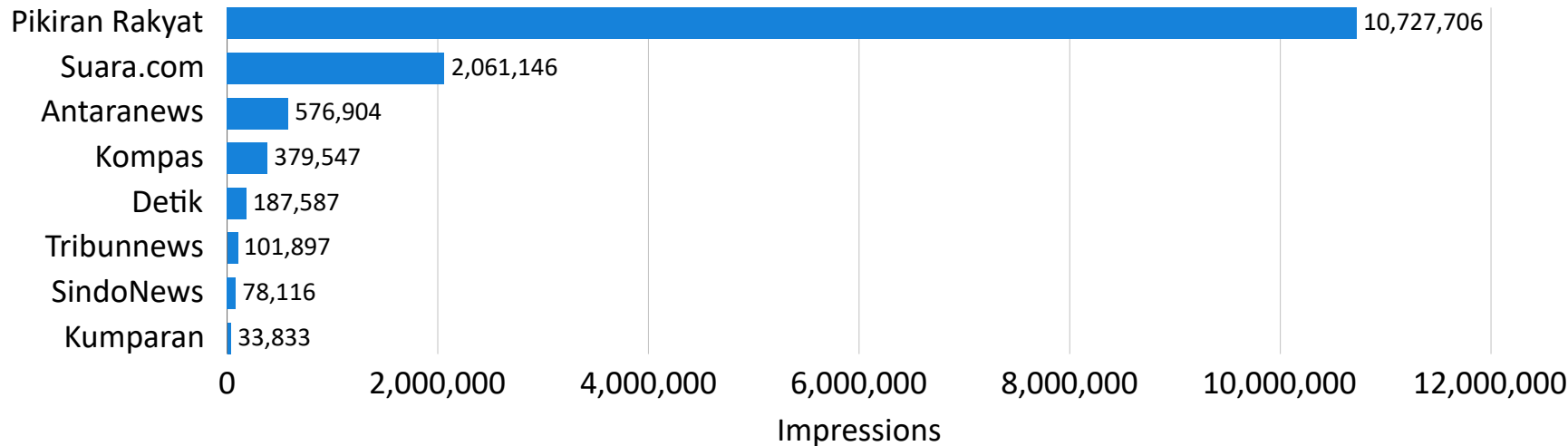
Total Estimated Revenue by Profile



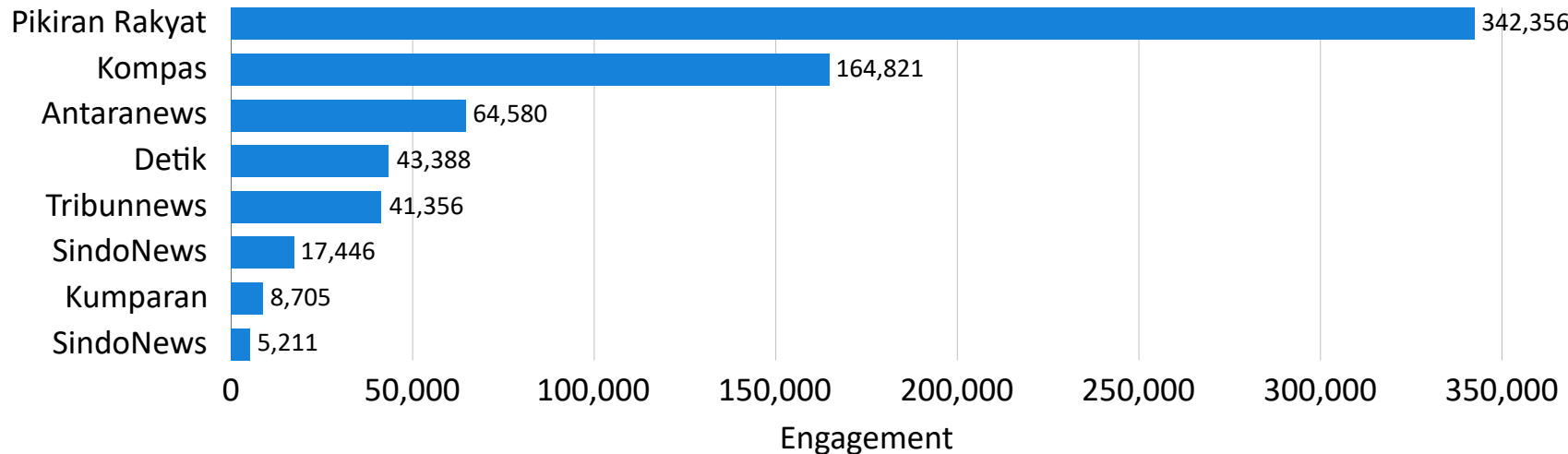
Total Reach by Profile



Total Impressions by Profile



Total Engagement by Profile



Key Insights

Pikiran Rakyat generated the highest **estimated revenue**, substantially outperforming all competitors.

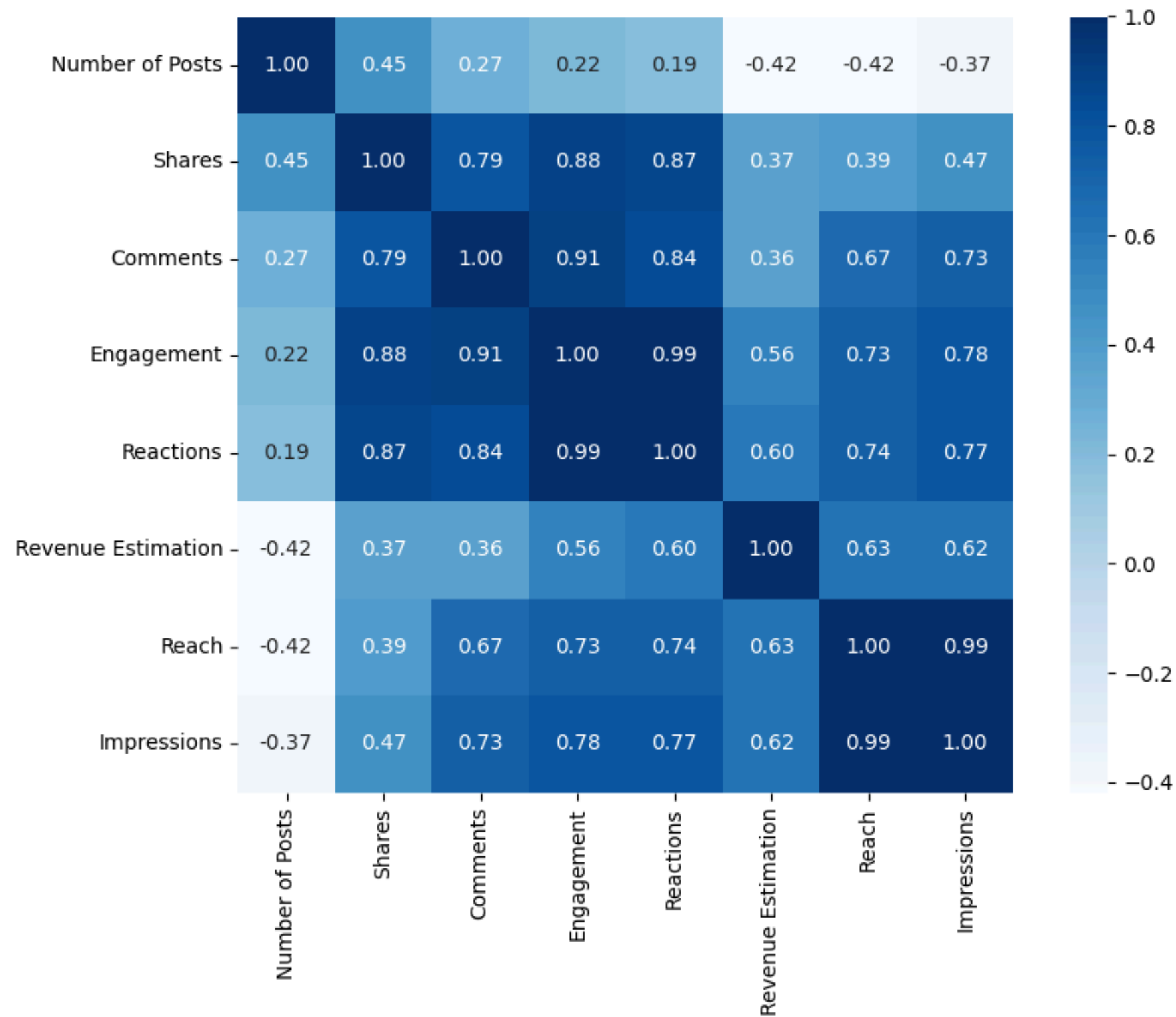
Pikiran Rakyat also recorded the highest **Reach and Impressions**, indicating the broadest content distribution.

Suara.com achieved the highest **Engagement**, but its estimated revenue remained considerably lower than Pikiran Rakyat.

The results suggested that audience engagement alone did not directly translate into **higher monetization revenue**.

Daily metrics correlation heatmap to understand relationships between content performance and estimated revenue.

Daily Metrics Correlation Heatmap

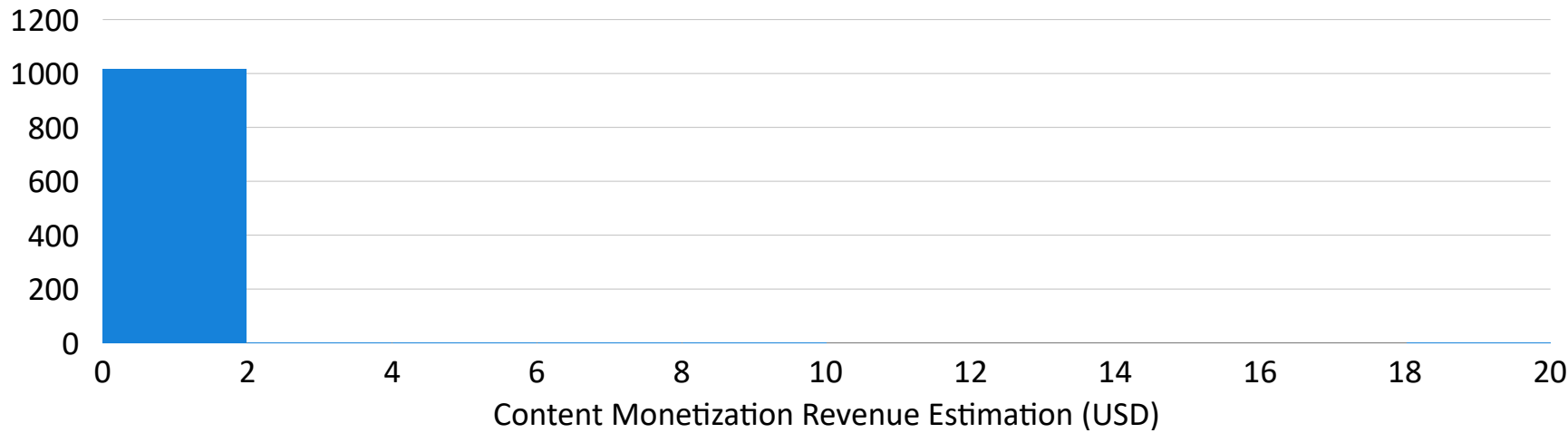


Daily Metrics Correlation Heatmap

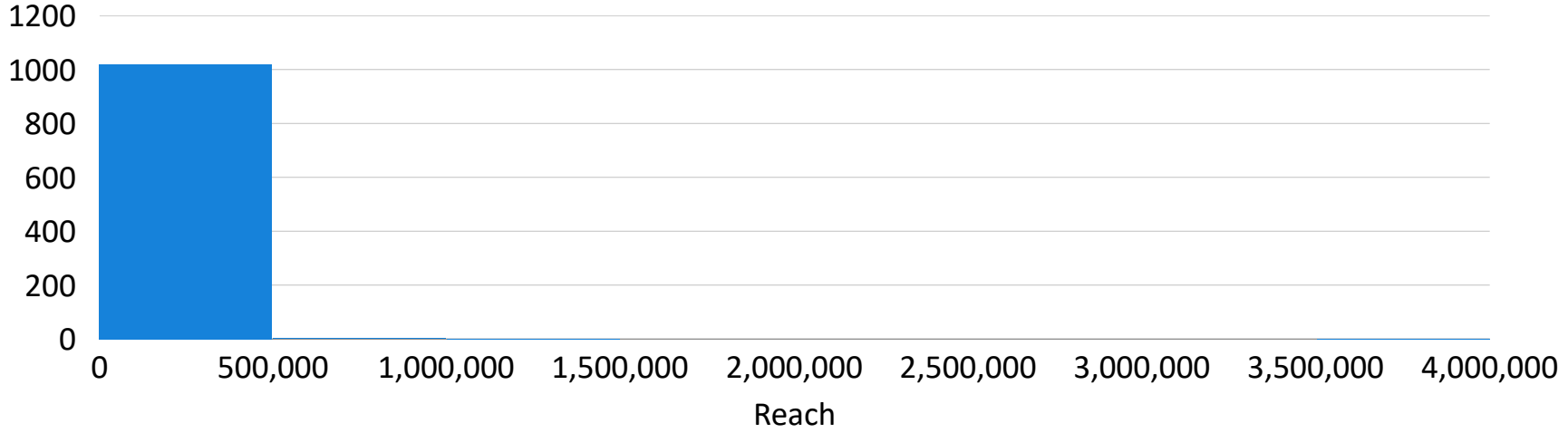
-  Revenue Estimation had a strong positive correlation with Reach (0.63) and Impressions (0.62), indicating that greater content distribution drove higher monetization potential.
-  Engagement-related metrics showed a moderate positive correlation with Revenue Estimation (0.36–0.60), with Reactions (0.60) having the highest correlation, meaning audience interactions supported monetization, though to a lesser extent than Reach and Impressions.
-  Number of Posts had a negative correlation with the most metrics, including Revenue (-0.42), indicating that posting more content did not necessarily lead to better daily performance or monetization.

Examining the distribution of key performance metrics before interpretation and further analysis.

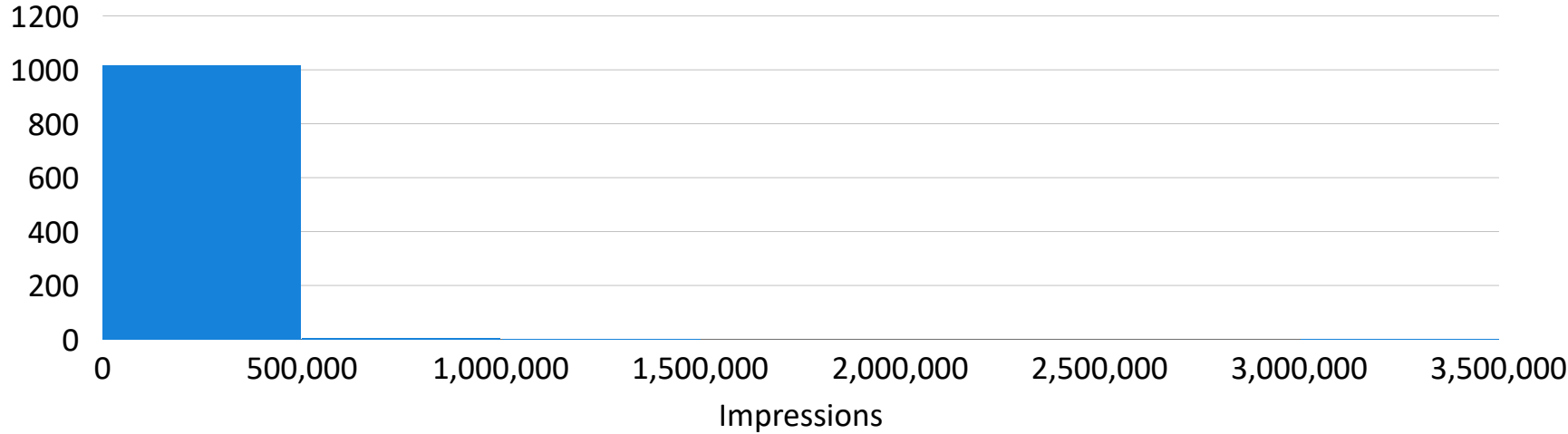
Estimated Revenue Distribution



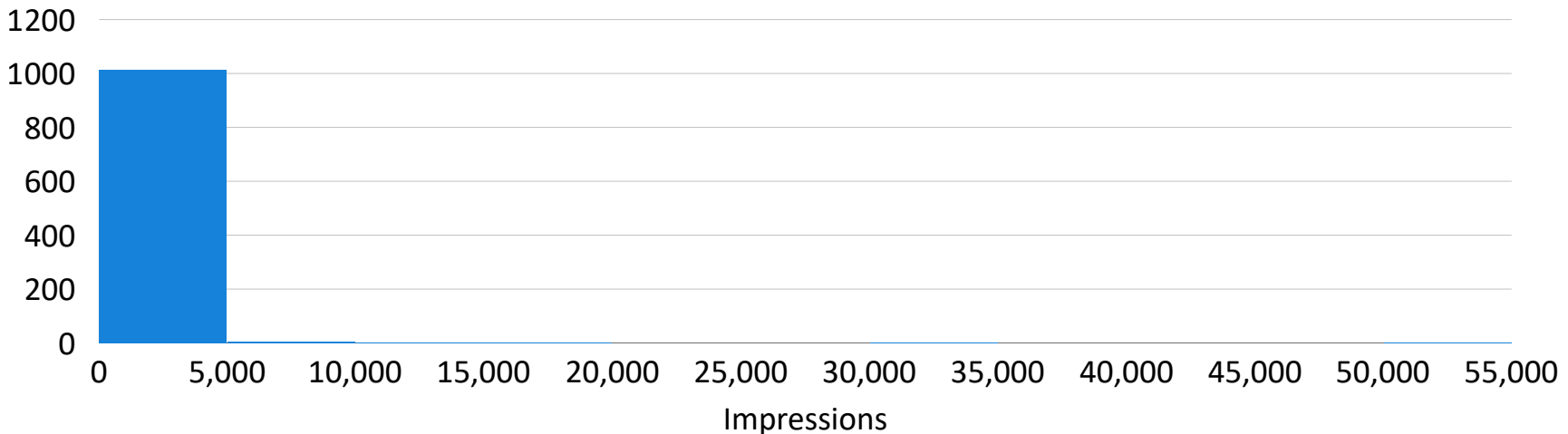
Reach Distribution



Impressions Distribution



Engagement Distribution



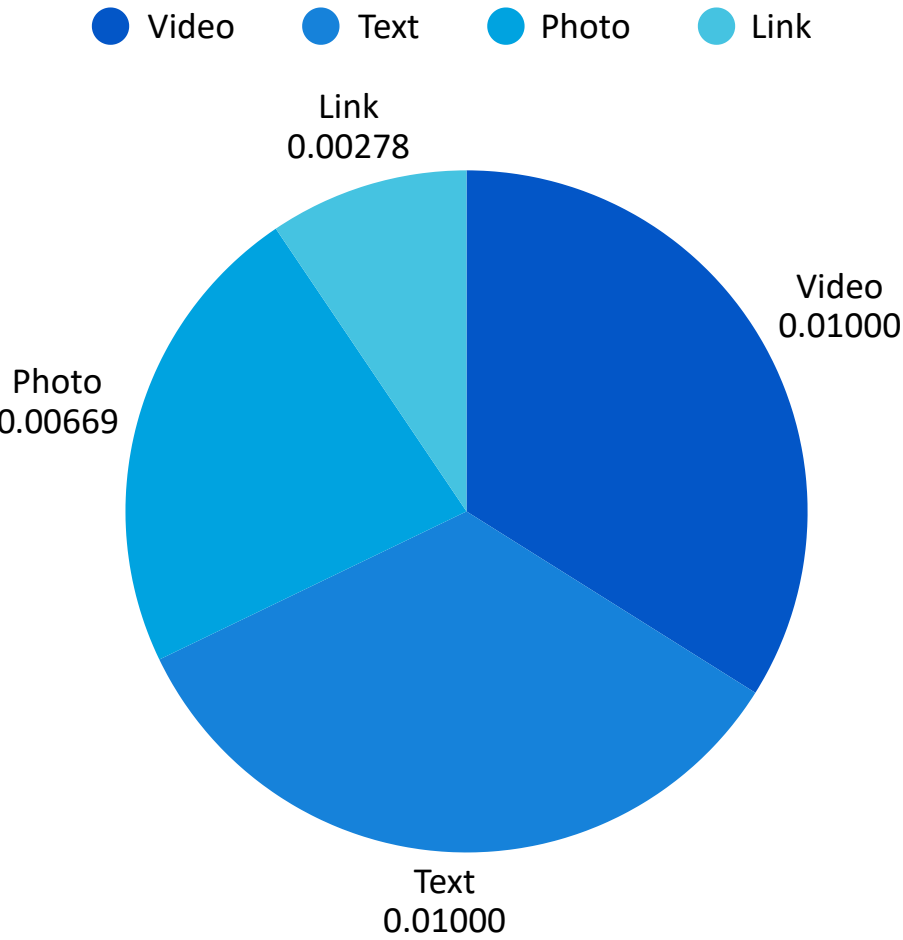
Key Insights

- ✓ Revenue, Reach, Impressions, and Engagement exhibit **positively skewed distributions**.
- ✓ Most posts have relatively low values, while a small number of posts generate exceptionally high results (**long-tail pattern**)
- ✓ The presence of **extreme outliers** indicates that the mean can be heavily influenced by a few high-performing posts.
- ✓ To ensure more robust and representative analysis, **median values** will be used for further analysis.

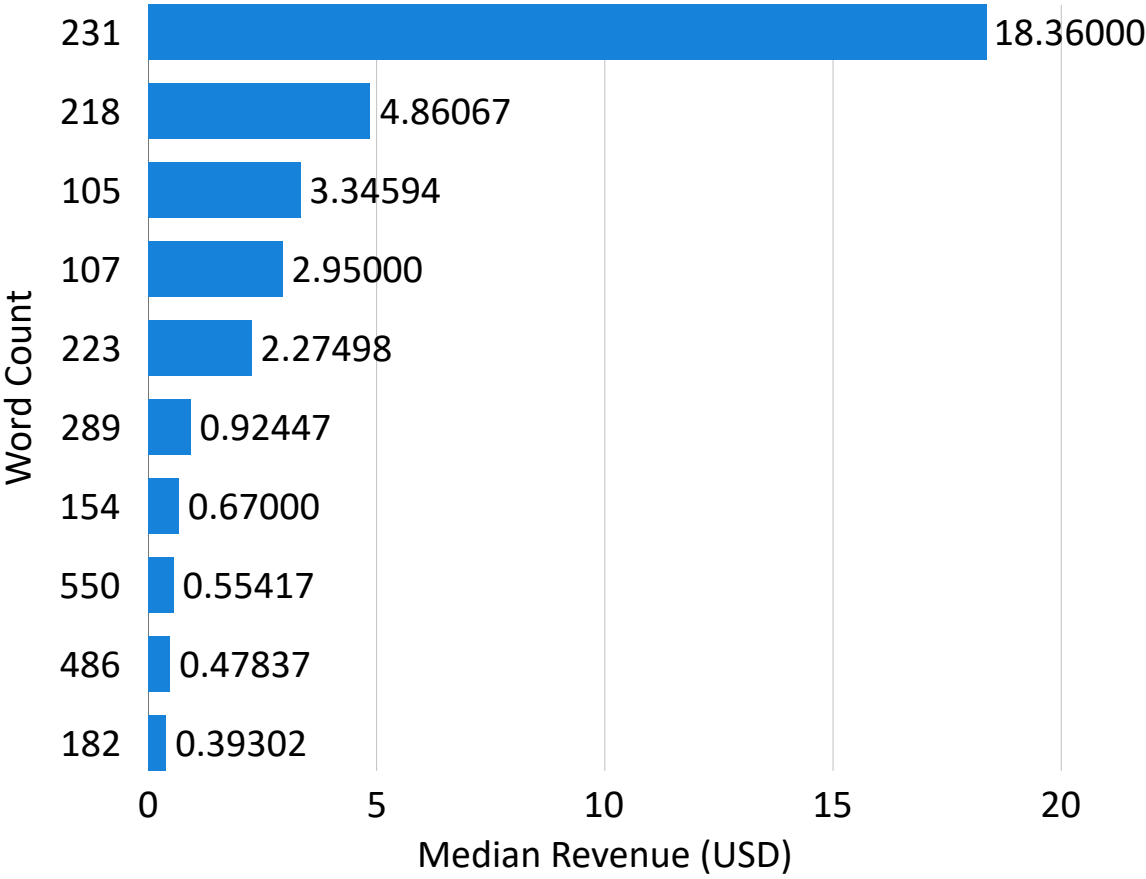
EDA: Revenue Performance by Content Characteristics

Analyzing how content type, caption length, and hashtag usage influenced median estimated revenue

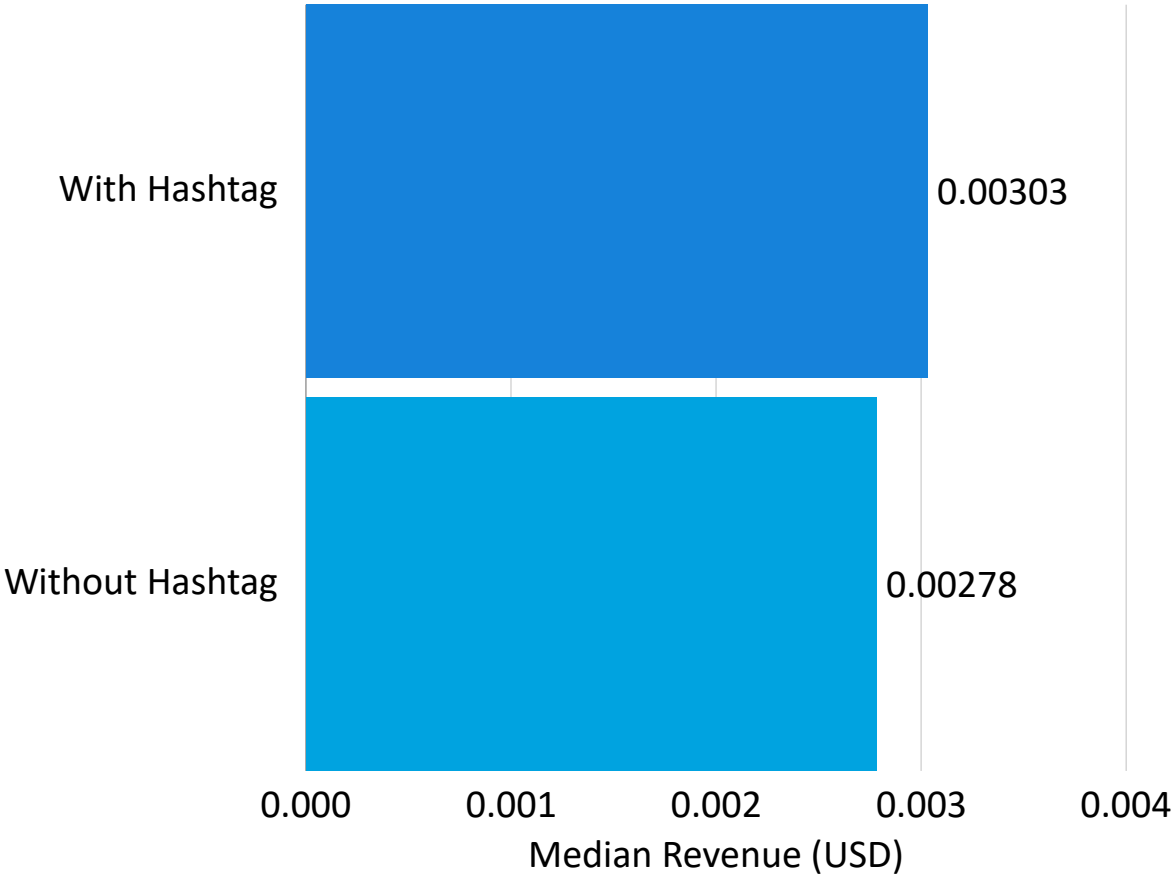
Median Revenue by Content Type



Median Revenue by Caption Length



Median Revenue by Hashtag Presence



Video and Text generated the highest median revenue, contributing more than 83% of the total median revenue across content types.



Median revenue was highest for caption with around 231 words and generally declined as they moved away from this range.



Posts with hashtag yield a slightly higher median revenue compared to posts without hashtags.



Key Insights



Video and Text deliver the highest median revenue, significantly outperforming Photos and Links.



Captions with around **231 words** achieved the highest median revenue, while very short or very long captions perform less.

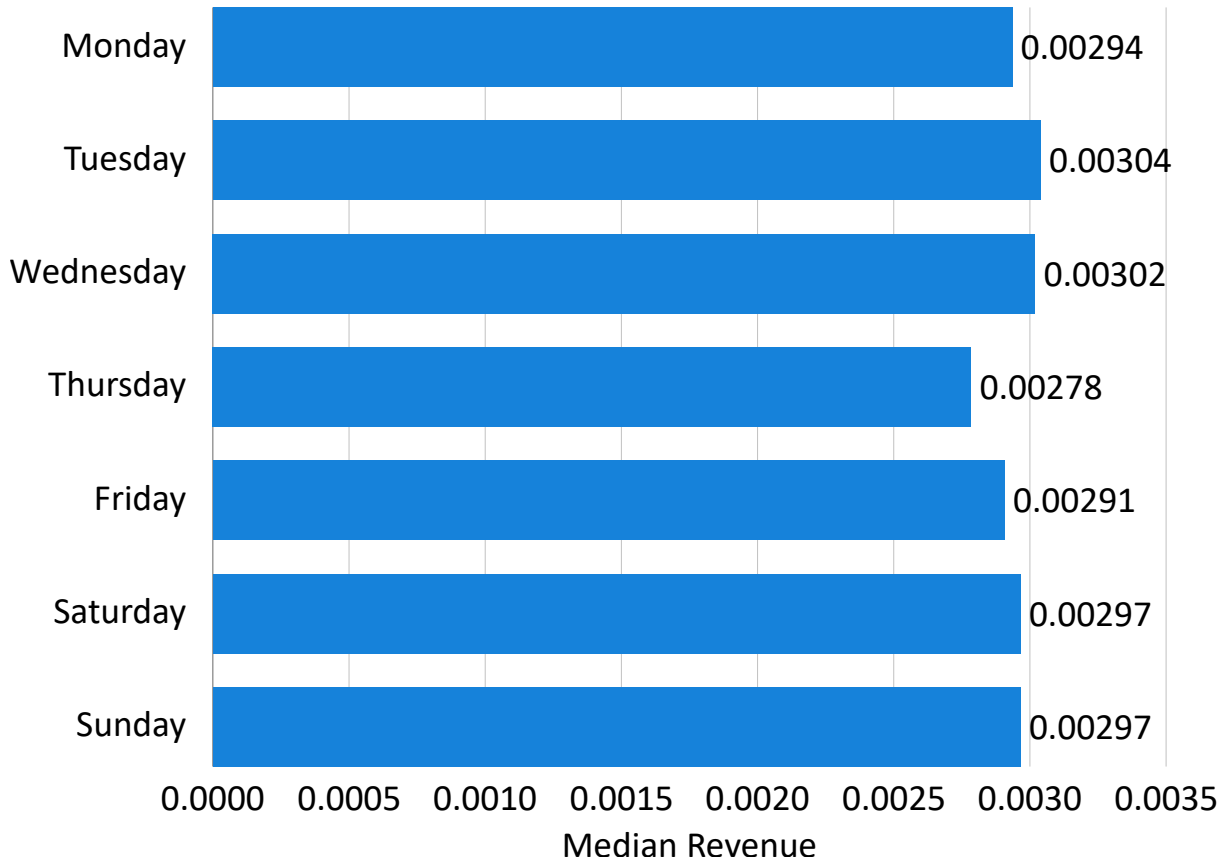


Using hashtags provides a small, but positive uplift in median revenue.



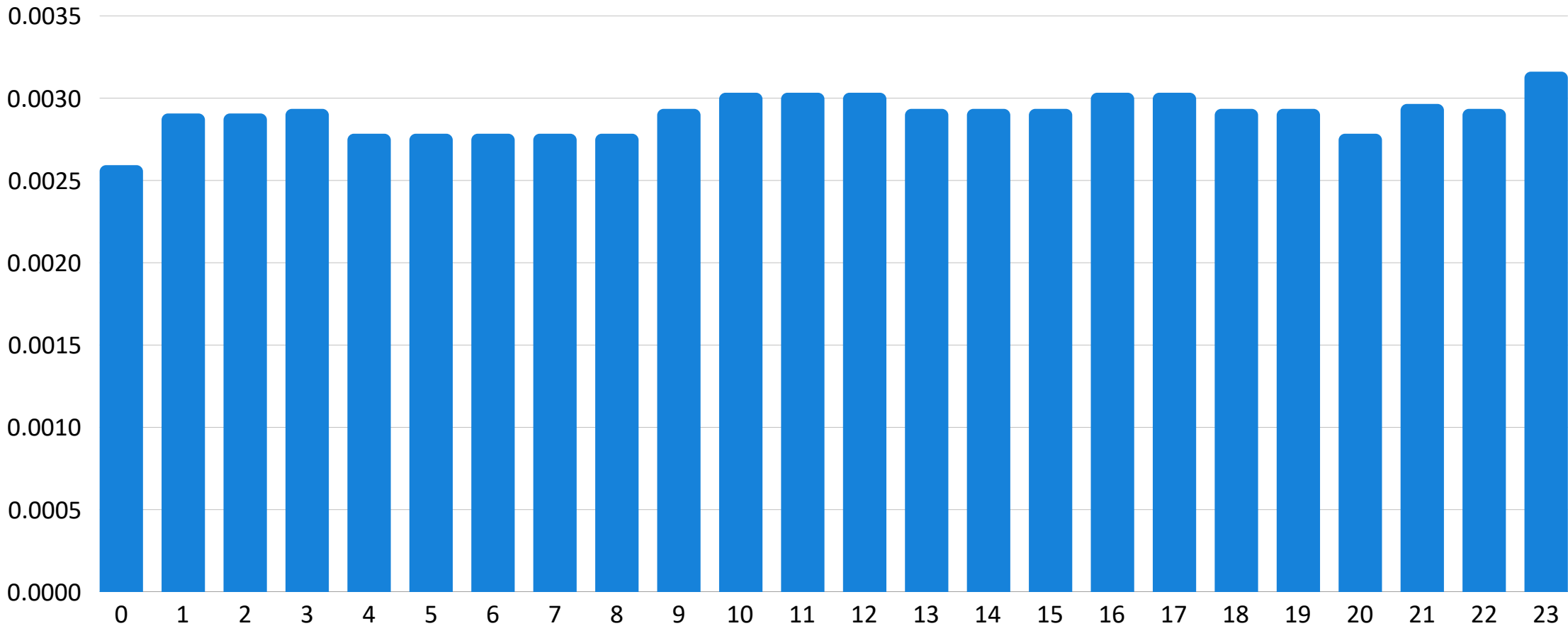
Content format, caption length, and hashtag usage also had roles in monetization performance, though to a smaller extent.

Median Revenue by Posting Day



Tuesday showed the highest median revenue (0.003039 USD), while Thursday was the lowest (0.002784 USD). Overall differences across days were relatively small.

Median Revenue by Posting Hour



Median revenue was relatively stable throughout the day, ranging from 0.002593 USD at 12:00 AM to 0.003161 USD at 11 PM. Late evening (11:00 PM) recorded the highest median revenue.



Key Insights



Posting performance was **relatively consistent** across days of the week.



Tuesday tended to yield the highest median revenue, while Thursday performed the lowest.



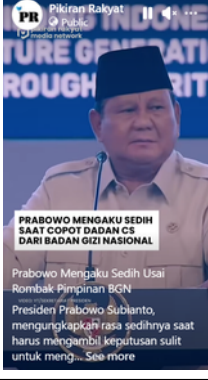
Revenue remains stable across most posting hours, with the best range for posting was **10 AM–12 PM**.



While timing has some impact, **content quality and other factors likely play a larger role in revenue**.

EDA: Top 3 Best Performing Content by Estimated Revenue

Highlighting the top 3 posts that generated the highest estimated revenue

Rank	Preview	Content Overview	Post info	Engagement Metrics	Reach & Impressions	Estimated Revenue (USD)
1		"Jeddah Fashion Week" Ala Jemaah Haji Makassar Lenggak-lenggok parade busana jemaah haji perempuan asal Makassar Sulawesi Selatan mencuri perhatian di Bandara Internasional King Abdulazis Jeddah Video	 6 June 2026 Saturday  00:33  231 words  0	 Reactions 4,010  Comments 2,107  Shares 423  Engagement 6,540	 Impressions 747,366  Reach 755,922	18.36 \$
2		Tinjau MBG di SMPN 111 Jakarta Barat, Prabowo: Makannya Enak Gak? Presiden Prabowo Subianto meninjau pelaksanaan Program Makan Gratis (MBG) di SMP Negeri 111 Jakarta Barat, Selasa, 2 Juni 2026. Video	 2 June 2026 Tuesday  05:36  218 words  4	 Reactions 40,864  Comments 12,032  Shares 735  Engagement 53,631	 Impressions 3,273,767  Reach 3,509,005	9.70 \$
3		Prabowo Mengaku Sedih Usai Rombak Pimpinan BGN Presiden Prabowo Subianto, mengungkapkan rasa sedihnya saat harus mengambil keputusan sulit untuk mengganti jajaran pimpinan di Badan Gizi Nasional (BGN). Video	 3 June 2026 Wednesday  08:53  105 words  4	 Reactions 8,809  Comments 6,820  Shares 394  Engagement 16,023	 Impressions 537,734  Reach 557,736	6.69 \$


Key Insights



All top 3 performing posts were **Videos**, confirming that this format drives higher monetization potential.



High engagement combined with **broad reach and impressions** consistently leads to higher revenue.



News and Human Interest topics (Hajj stories, government programs, investigation) resonate strongly with audience.



All top 3 posts were from **Pikiran Rakyat** accounts, confirming the dominance in estimated revenue than competitors.

Recommendations	Key Finding	Recommended Action	Expected Impact
 Prioritize Video Content	Video content generated the highest median revenue and dominated the top performing posts.	Increase the proportion of video content, especially on high interest topics, such as News and Human Interest.	Higher Revenue, Reach, Impressions, and Engagement.
 Strengthen Content Distribution	Reach and Impressions were the strongest drivers of revenue in the prediction model.	Improve content distribution through a better headlines, thumbnails, and promotions.	Broader audience exposure and higher revenue potential.
 Optimize Caption Strategy	Captions around 231 words and hashtag usage produced slightly higher median revenue.	Use informative captions with optimal length and relevant hashtags.	Better audience exposure and interaction also higher revenue potential.
 Prioritize Posting on Prime Day and Time	Posts published at Tuesday and 11:00 PM generated higher median revenue.	Prioritize posting at Tuesday and optimal hours, such as 11:00 PM or 10 AM–12 PM.	Higher revenue potential.
 Prioritize Content Quality over Quantity	Number of posts daily had negative correlation with revenue	Focus on producing content that can reach wider audiences and invite huge interactions rather than increasing posting frequency.	More effiecient resources allocation and sustainable monetization growth.

Conclusion



Developed a **machine learning model to estimate Facebook Content Monetization Revenue** using content characteristics, posting information, and engagement metrics.



XGBoost achieved the best performance (**$R^2 = 0.7018$, $RMSE = 0.2744$, and $MAE = 0.0728$**) and was selected to estimate missing revenue values.



Reach, Impressions, and engagement-related metrics were identified as the primary drivers of monetization revenue, while content type, caption characteristics, and posting time provided additional, but smaller contributions.

Limitations



The model was trained using only **one month** historical data, limiting its ability to capture longer-term seasonal trends.



Several potentially important features contained substantial **missing values or were unavailable**, limiting the model's predictive performance.



The model estimated revenue based on **observed post-performance metrics** rather than forecasting revenue before content is published.

Next Steps



Collect **additional historical data** covering multiple months or years to improve model robustness.



Incorporate **richer features** such as demographics, duration, topic classification, clicks, sentiment, etc.



Perform **hyperparameter optimization** and evaluate **additional regression models** to improve prediction.

LET'S CONNECT



<https://www.linkedin.com/in/muhammad-farel-daffa>



fareldf4@gmail.com



+6281211999109